
Segment-Aware Pricing: Does Accident History Penalize Premium Vehicles More Than Budget Ones?

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Abstract

The used car market presents significant pricing opacity for consumers, dealers, and insurers alike. Existing price prediction models treat accident history as a uniform depreciating factor, ignoring potential heterogeneity in its impact across vehicle market segments. In this work, we propose a segment-aware pricing framework that integrates unsupervised K-Means clustering with supervised regression models to investigate whether accident history penalizes premium vehicles disproportionately compared to budget vehicles. Using a dataset of 4,009 used car listings scraped from cars.com, we perform rigorous data preprocessing including feature extraction, binary mapping, one-hot encoding, and outlier removal. K-Means clustering ($k = 3$) identifies three distinct market segments: Budget, Mid-Range, and Premium, validated by Silhouette Score (0.3525) and the Elbow Method. We train Linear Regression, Random Forest, and XGBoost models both globally and per-segment, evaluating performance with RMSE and R^2 . Our key finding is that accident history does not penalize uniformly: the Mid-Range segment sustains the largest proportional penalty (15.9%), while the Premium segment sustains the largest absolute penalty (\$10,161). SHAP analysis reveals that the accident feature exhibits its widest impact distribution in the Premium segment, indicating greater pricing unpredictability for accident-damaged premium vehicles. These findings have direct implications for used car buyers, dealers, and insurance companies.

1 Introduction

The United States used car market represents one of the largest consumer markets in the country, with annual transaction volumes in the hundreds of billions of dollars. Despite its scale, pricing in this market remains highly opaque. A buyer comparing two similar vehicles with different accident histories has limited tools to quantify the actual financial impact of that history, and sellers face symmetric uncertainty in setting fair prices.

Traditional approaches to used car price prediction rely on supervised regression models that treat all vehicles uniformly, learning a single global relationship between vehicle attributes and price [3, 2]. While such models achieve reasonable predictive performance, they obscure an important economic reality: the same attribute, such as accident history, may carry vastly different pricing implications depending on the segment of the market in which the vehicle competes.

This asymmetry matters for three distinct stakeholders. First, buyers in the premium segment face larger absolute financial exposure when purchasing an accident-damaged vehicle, yet lack quantitative guidance on the size of that exposure. Second, dealerships and private sellers need segment-aware valuation tools to price trade-ins accurately, particularly for post-accident vehicles. Third, insurance companies computing diminished value claims after accidents must account for segment-dependent depreciation rather than applying a flat percentage reduction.

In this project, we propose a segment-aware pricing framework that combines unsupervised vehicle segmentation with supervised price modeling to answer the following research question: Does

accident history penalize premium vehicles disproportionately compared to budget vehicles, and does a segment-aware model better capture these dynamics than a global baseline?

Our contributions are as follows:

- We apply K-Means clustering to identify three data-driven vehicle market segments (Budget, Mid-Range, Premium) without relying on arbitrary brand-based categorization.
- We compare global and segment-specific XGBoost regression models, revealing that segment-specific models yield lower absolute RMSE within each segment despite the global model explaining more overall variance.
- We quantify the accident price penalty per segment in both absolute dollar and percentage terms, revealing a non-uniform penalty structure that challenges the assumption of uniform depreciation.
- We apply SHAP analysis to demonstrate that the accident feature exhibits segment-dependent impact distributions, with Premium vehicles showing the highest variance in accident-driven price penalty.

2 Related Work

Prior research on used car price prediction and market segmentation falls into three broad categories: supervised price prediction, clustering-based market segmentation, and hybrid interpretable frameworks.

2.1 Used Car Price Prediction

Early work in used car price prediction applied multiple linear regression to identify key pricing factors. Sun (2024) analyzed a dataset from the same cars.com source used in this study, finding that mileage and model year are the dominant price drivers, while color has negligible statistical effect [2]. Importantly, that work excluded accident history as a feature by filtering out accident-reported vehicles entirely, leaving its pricing impact unstudied. Hankar et al. (2022) compared multiple regression techniques on a used car dataset collected from an online marketplace, finding that gradient boosting regressors consistently outperform linear baselines, achieving R^2 scores of approximately 0.80 [3]. Our work extends these baselines by conditioning price models on market segment, allowing feature importance to vary across vehicle tiers.

2.2 Clustering for Market Segmentation

K-Means clustering has a long history as a tool for market segmentation. Kuo et al. (2002) proposed a two-stage framework integrating self-organizing feature maps with K-Means for consumer market segmentation, demonstrating that data-driven segmentation outperforms arbitrary categorical groupings [4]. Their finding that the combination of unsupervised clustering followed by supervised modeling yields superior interpretability motivates our hybrid approach. Unlike brand-based segmentation used in prior automotive studies, our clustering operates on continuous vehicle attributes (price, mileage, horsepower, age), producing segments that reflect actual market behavior rather than manufacturer positioning.

2.3 Interpretability and Hybrid Frameworks

Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), a unified framework for interpreting model predictions grounded in cooperative game theory [1]. SHAP assigns each feature a contribution value for each individual prediction, enabling both global feature ranking and local explanation. In our work, SHAP is used to decompose the accident feature’s impact distribution across segments, providing fine-grained evidence for the segment-dependent penalty hypothesis beyond what aggregate statistics alone can reveal.

Gap addressed: While prior work has studied used car price prediction and market segmentation independently, no published study has combined data-driven vehicle segmentation with a systematic analysis of how accident history’s pricing impact varies across those segments. This work fills that gap.

3 Methodology

3.1 Dataset

The dataset consists of 4,009 used car listings collected from cars.com and made publicly available on Kaggle. Each listing contains 12 attributes: brand, model, model year, mileage, fuel type, engine description, transmission, exterior color, interior color, accident history, clean title status, and listing price. The accident history column contains two values: “None reported” and “At least 1 accident or damage reported,” making it a binary feature after encoding.

3.2 Data Preprocessing

We applied a comprehensive preprocessing pipeline to prepare the data for modeling. All steps were performed in strict order to prevent data leakage.

Feature Engineering:

Listing price was cleaned by removing dollar signs and commas and converted to a float. Mileage was similarly cleaned of the “mi.” suffix. A horsepower feature was extracted from the engine description string using regular expression matching for the numeric HP value. Car age was derived as $2025 - \text{model_year}$, providing a more interpretable continuous feature than the raw year.

Binary Mapping:

The accident column was mapped to a binary integer (0 = none reported, 1 = at least one accident). The clean title column was mapped to 1 for “Yes” and 0 for missing values, encoding the absence of a clean title as a negative signal.

Missing Value Handling:

Fuel type (170 missing values) was imputed with the mode. Rows with missing accident or horsepower values were dropped rather than imputed, as these features are central to the research question and clustering pipeline respectively. This produced 3,054 records after additional price outlier removal.

Outlier Removal:

Price outliers beyond the 1st and 99th percentiles were removed. Inspection revealed implausible listings (e.g., a Maserati Quattroporte at \$2,954,083), likely data entry errors. After removal, the price range was \$3,999 to \$179,900.

One Hot Encoding:

Categorical features: brand, fuel type, transmission, exterior color, and interior color: were one-hot encoded with `drop_first=True` to avoid multicollinearity, expanding the feature space to 109 columns.

Train-Test Split and Scaling:

Data was split 80/20 before any scaling. `StandardScaler` was fit exclusively on the training set and applied to both sets, ensuring no leakage of test distribution information into preprocessing.

3.3 Customer Segmentation via K-Means

K-Means clustering was applied to a four-feature subspace: price, mileage, horsepower, and car age. To select the optimal k , we evaluated $k \in \{2, \dots, 10\}$ using the Elbow Method (inertia) and Silhouette Score. The Elbow Method showed the steepest inertia drop between $k = 2$ and $k = 3$. The Silhouette Score peaked at $k = 3$ (0.3525), confirming three clusters as the most compact and well-separated partition. K-Means minimizes the within-cluster sum of squares:

$$\arg \min_S \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \quad (1)$$

where S_i are the cluster assignments and $\boldsymbol{\mu}_i$ are cluster centroids.

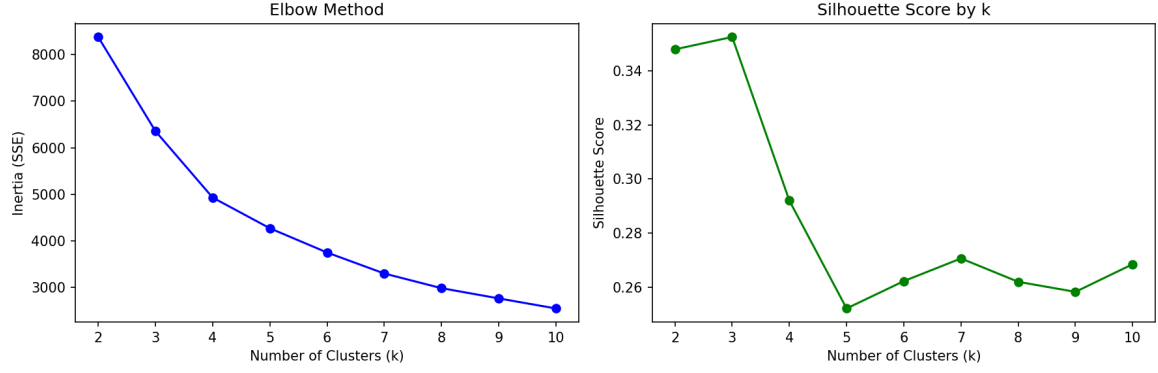


Figure 1: Elbow Method (left) and Silhouette Score (right) for $k = 2$ to 10. Both criteria support $k = 3$ as the optimal number of clusters.

The three clusters were labeled by average price: Budget (avg. \$14,305, 122k miles, 273HP, 16 years old), Mid-Range (avg. \$31,819, 53k miles, 300HP, 8 years old), and Premium (avg. \$75,150, 27k miles, 480HP, 6 years old).

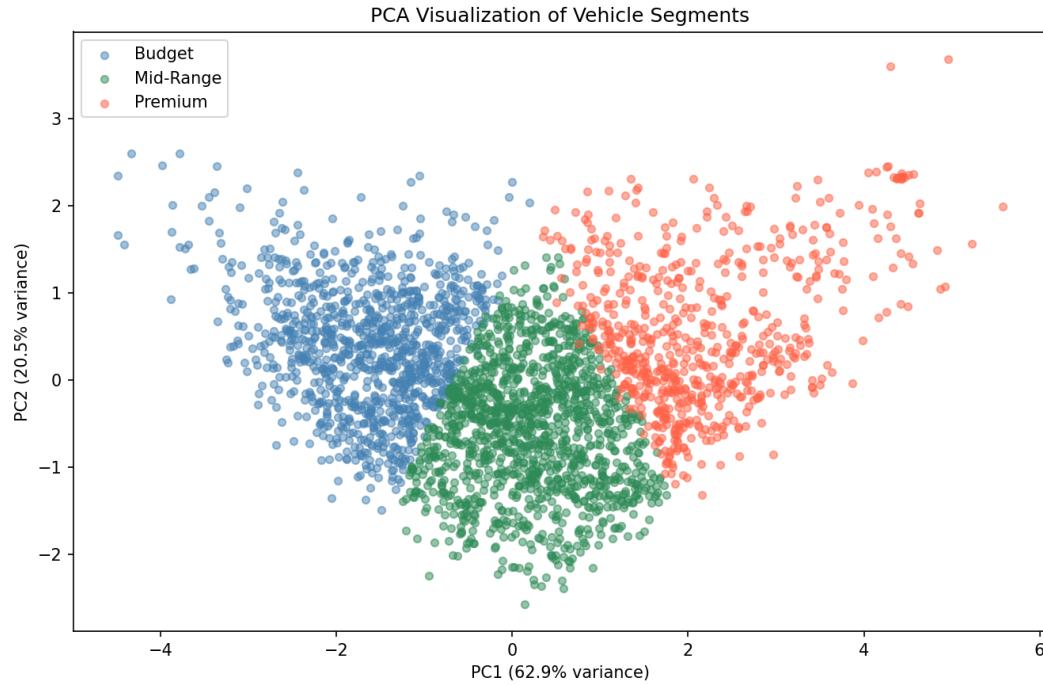


Figure 2: PCA 2D projection of vehicle clusters. Budget (blue), Mid-Range (green), and Premium (red) segments show clear spatial separation. Two principal components explain 83.4% of total variance.

3.4 Supervised Regression Models

We trained three supervised regression models. **Linear Regression** served as a global baseline:

$$\hat{y} = \beta_0 + \sum_{i=1}^p \beta_i x_i \quad (2)$$

Random Forest [6] is an ensemble of decision trees built via bootstrap sampling, reducing variance relative to a single tree while providing feature importance scores.

XGBoost [5] trains trees sequentially with regularization. The objective function is:

$$\text{Obj} = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{m=1}^M \Omega(h_m) \quad (3)$$

where L is MSE loss and Ω penalizes model complexity. XGBoost was trained both as a global model across all vehicles and as three separate segment-specific models.

3.5 SHAP Interpretability

SHAP values [1] were computed for the global XGBoost model using TreeExplainer. We use SHAP in two ways: globally to rank all 108 features by mean absolute SHAP value, and locally to analyze the distribution of SHAP values for the accident feature across the three segments.

4 Experiments and Results

4.1 Global Model Performance

Table 1 summarizes model performance on the held-out test set.

Table 1: Performance comparison of all models on the test set.

Model	RMSE	R ²
Linear Regression (Global)	\$16,124	0.6954
Random Forest (Global)	\$12,591	0.8142
XGBoost (Global)	\$10,830	0.8626
XGBoost (Budget)	\$5,010	0.5316
XGBoost (Mid-Range)	\$6,840	0.6505
XGBoost (Premium)	\$17,819	0.5971

The global XGBoost model achieves the best overall performance (RMSE = \$10,830, R² = 0.863). Segment-specific models yield lower absolute RMSE for Budget and Mid-Range, reflecting narrower price ranges within those segments. However, lower R² scores for segment models indicate that cross-segment patterns contribute to global predictive accuracy, and that segment-specific models trained on smaller subsets sacrifice some explanatory power.

4.2 Accident Penalty Analysis

Table 2 shows the mean price difference between accident-reported and clean vehicles within each segment.

Table 2: Accident price penalty by market segment.

Segment	No Accident	Has Accident	Penalty (\$)	Penalty (%)
Budget	\$14,754	\$13,620	\$1,134	7.7%
Mid-Range	\$33,225	\$27,956	\$5,270	15.9%
Premium	\$76,566	\$66,405	\$10,161	13.3%

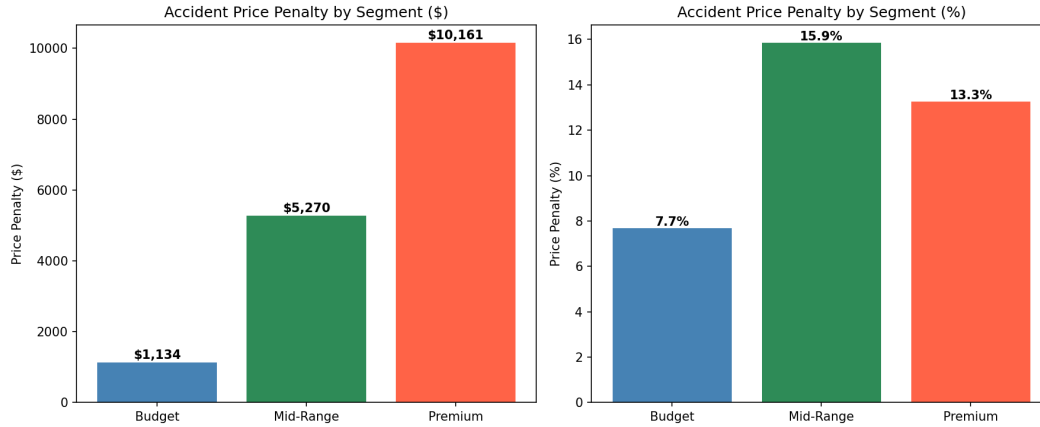


Figure 3: Accident price penalty by segment in absolute dollars (left) and percentage (right). Mid-Range vehicles sustain the largest proportional penalty while Premium vehicles sustain the largest absolute penalty.

The results reveal a non-uniform accident penalty structure. Budget vehicles sustain the smallest proportional penalty (7.7%), consistent with the interpretation that buyers in this segment expect some degree of wear and price accordingly. Mid-Range vehicles sustain the largest proportional penalty (15.9%), suggesting heightened buyer sensitivity to accident history relative to vehicle value. Premium vehicles sustain the largest absolute dollar penalty (\$10,161) but a lower proportional penalty (13.3%) than Mid-Range, indicating that premium buyers discount accident history less aggressively in percentage terms.

4.3 SHAP Feature Importance

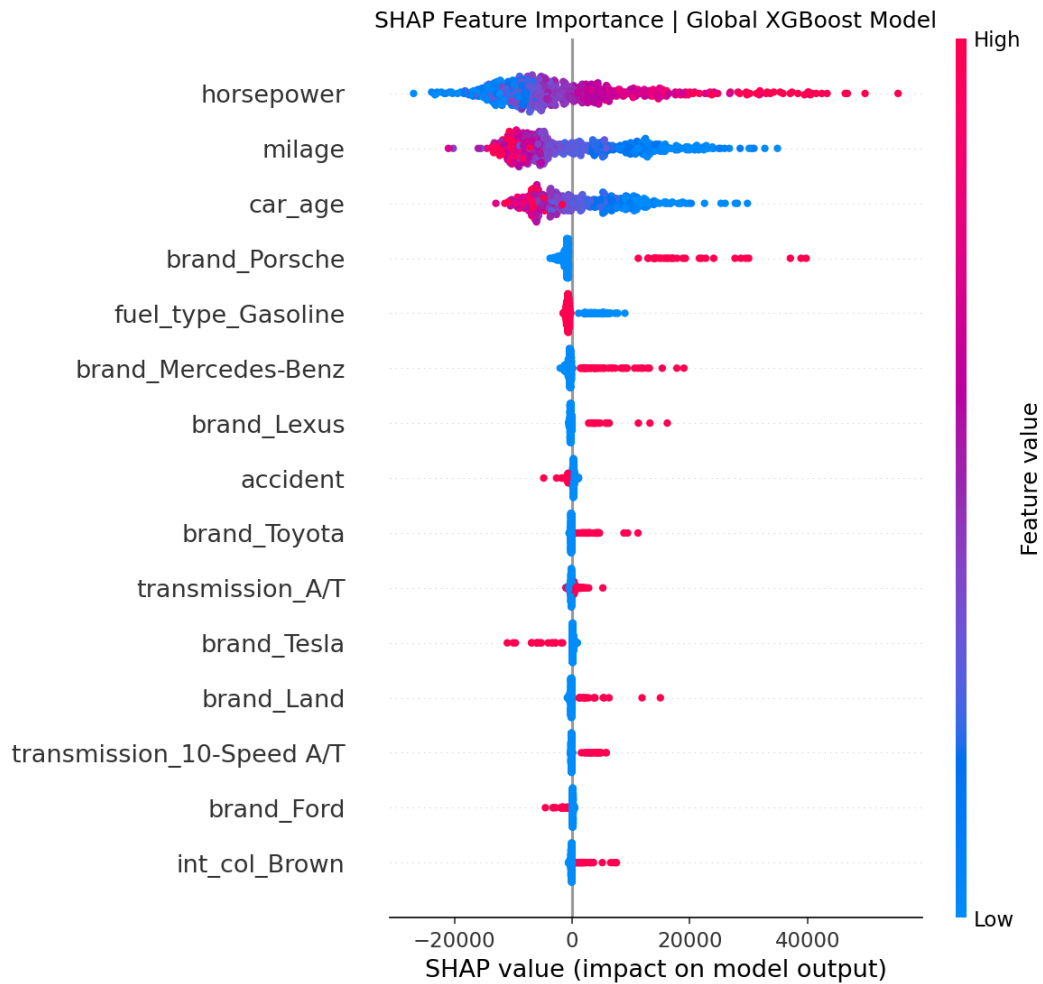


Figure 4: SHAP summary plot for the global XGBoost model. Horsepower, mileage, and car age are the dominant price drivers. The accident feature ranks 8th, with consistently negative SHAP values confirming it as a price-reducing signal.

The SHAP summary plot confirms that horsepower is the dominant price driver, followed by mileage and car age. The accident feature ranks 8th out of 109 features by mean absolute SHAP value, with its has-accident value (pink) consistently pushing predicted prices lower.

4.4 Segment-Wise SHAP Accident Distribution

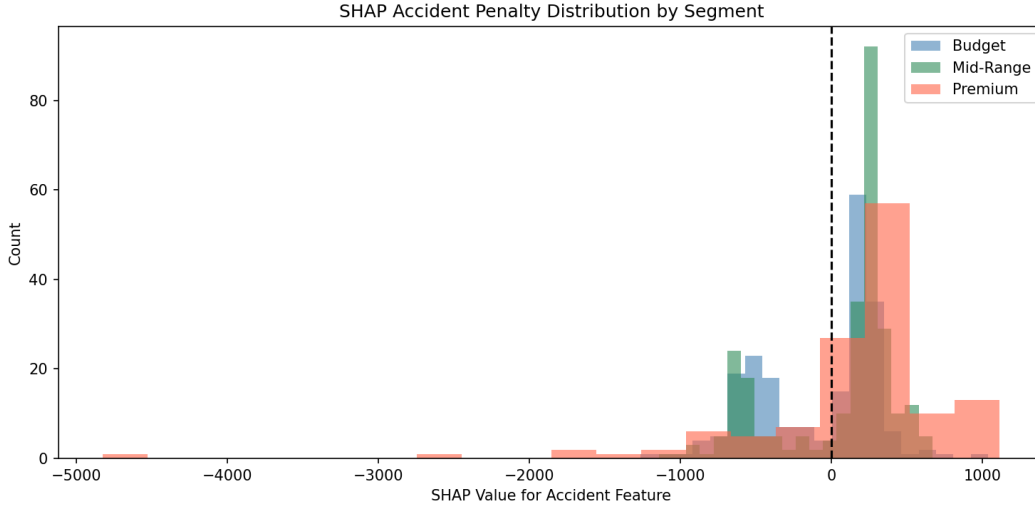


Figure 5: Distribution of SHAP values for the accident feature by segment. Budget and Mid-Range vehicles cluster near zero, while Premium vehicles exhibit a wide spread with a long negative tail, indicating high variance in accident penalty for premium vehicles.

Budget and Mid-Range vehicles cluster tightly near zero SHAP values for the accident feature, indicating that for most vehicles in these segments, the model attributes a small contribution to accident history. Premium vehicles show the widest distribution, with a long tail extending to approximately $-\$5,000$, indicating that some premium vehicles sustain very large model-attributed penalties while others are barely affected. This variance suggests that accident history interacts differently with other features in the Premium segment.

5 Conclusion

This study proposed a segment-aware pricing framework for used vehicles that integrates K-Means clustering with XGBoost regression to investigate whether accident history penalizes vehicles differently across market segments. Using 4,009 US used car listings, we identified three data-driven segments and quantified the accident price penalty within each.

Our primary finding is that the accident penalty is non-uniform. Mid-Range vehicles sustain the largest proportional penalty (15.9%), Premium vehicles sustain the largest absolute penalty (\$10,161) with the highest variance per SHAP analysis, and Budget vehicles are least affected proportionally (7.7%). These findings have practical implications: insurance diminished value claims should be segment-conditioned, and dealership pricing tools should incorporate segment-aware accident adjustments.

Limitations. Accident history is binary in this dataset, capturing only presence or absence without severity. A minor fender-bender and a structural collision are treated identically, which likely suppresses the true magnitude of the penalty for serious accidents. The dataset is limited to a single platform (cars.com) and may not generalize across all US used car transactions. Segment-specific models yield lower R^2 than the global model, reflecting the cost of smaller training subsets.

Future Work. Future directions include sourcing datasets with accident severity ratings to test whether severity modulates segment-dependent penalty, and temporal analysis examining whether the penalty structure shifted during the COVID-19 used car market anomaly.

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